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Inventor(s)

Frischen; Andreas et al.

Method for Generating a Training Dataset

Abstract

A computer-implemented method for generating a training dataset for training an artificial intelligence for operating a measuring device, in particular a wall diagnostic device, includes (i) receiving unclassified sensor data of at least one sensor unit of a measuring device by a classifier module, (ii) classifying the unclassified sensor data and providing classified sensor data using a classifier module, and (iii) adding the classified sensor data to a training dataset.

Inventors: Frischen; Andreas (Leonberg, DE), Koelsch; Andreas (Peiting, DE), Giliard; Anke (Stuttgart, DE), Moster; Benjamin (Niefern-Oeschelbronn, DE), Sgarz; Heiko (Leonberg, DE), Steuler; Jan Linus (Woelfersheim, DE), Law; Kakit (Reutlingen, DE), Fiederer; Lukas (Balingen, DE), Dekel; Tal (Ramat Gan, IL), Troeltzsch; Uwe (Roesrath, DE)

Applicant: Robert Bosch GmbH (Stuttgart, DE)

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Background/Summary

[0001] This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2024 202 245.4, filed on Mar. 11, 2024 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

[0002] The present disclosure relates to a method for generating a training dataset for training an artificial intelligence for a wall diagnostic device.

BACKGROUND

[0003] Diagnostic devices for performing wall diagnostics and detecting objects formed in the walls are known from prior art.

[0004] It is a task of the present disclosure to provide an improved method for generating a training dataset for training an artificial intelligence of a wall diagnostic device.

[0005] The task is solved by the method set forth below. Advantageous embodiments are subject-matter also set forth below.

SUMMARY

[0006] According to one aspect, a computer-implemented method is provided for generating a training dataset for training an artificial intelligence to operate a measuring device, in particular a wall diagnostic device, wherein the method comprises: [0007] receiving unclassified sensor data of at least one sensor unit of a measuring device via a classifier module, wherein the sensor data depicts a wall to be diagnosed having a wall type, and wherein an object of an object type is disposed in the wall in an object position; [0008] classifying the unclassified sensor data and providing classified sensor data via a classifier module, wherein the classifier module is configured as an artificial intelligence and trained via semi-supervised learning to determine a classification for the unclassified sensor data with respect to the wall type and/or object position and/or object type of the object; and adding the classified sensor data to a training dataset.

[0009] This may achieve the technical advantage of providing an improved method for generating a training dataset for training an artificial intelligence to operate a measuring device, in particular a wall diagnostic device. For this purpose, initially unclassified sensor data of at least one sensor unit of a measuring device is received by a classifier module. The sensor data depicts a wall of a wall type to be examined and objects disposed in the wall. The unclassified sensor data is classified via the classifier module and converted into correspondingly classified sensor data.

[0010] The classifier module is configured as an artificial intelligence trained via a semi-supervised learning process to find a corresponding classification for unclassified sensor data. The classification is thereby carried out in relation to the wall type and/or the object position and/or the object type of the objects disposed in the wall. The sensor data correspondingly classified by the classifier module, is subsequently added to the training dataset to be generated.

[0011] By using a correspondingly trained artificial intelligence as a classifier module, precise and reliable classification of unclassified sensor data can be carried out. The classified sensor data is labeled with respect to the classification. The classification states which object is positioned or present in the wall type of the wall depicted by the sensor data, or in which object position the object is disposed or states the object type of an object disposed in the wall.

[0012] Classified sensor data corresponds to the labeled sensor data known from the prior art of machine learning. The unclassified sensor data corresponds to the unlabeled sensor data known from the prior art. The classification of the classified sensor data thus corresponds to a labeling of the sensor data with respect to the particular classification feature.

[0013] According to one embodiment, the classifier module was trained on classified sensor data and sensor data with pseudo-classifications, in order to determine corresponding classifications for unclassified sensor data, wherein the pseudo-classifications were predicted by a pseudo-

classification module based on unclassified sensor data, and wherein the pseudo-classification module is configured as an artificial intelligence and is trained on classified sensor data to determine pseudo-classifications for unclassified sensor data.

[0014] This may achieve the technical advantage that an improved classifier module may be provided to classify unclassified sensor data. The classifier module is trained on a training dataset comprising classified sensor data and sensor data with a pseudo-classification. The sensor data with a pseudo-classification was in turn classified by a pseudo-classification module. The pseudo-classification module, in turn, is trained on classified sensor data to determine pseudo-classifications for unclassified sensor data. The pseudo-classifications are classifications of the sensor data with respect to the classification characteristics mentioned above that were generated by the pseudo-classification module.

[0015] The classified sensor data, on the other hand, is not automatically classified, but rather has been manually classified as is common in the prior art in the field of machine learning. The sensor data may be based on measurements from measuring devices carried out on real walls of buildings. Alternatively or additionally, the sensor data may be based on measurements of laboratory walls.

[0016] The laboratory walls have the advantage that all features of the wall or the objects disposed in the walls are known to the experimenter. This may result in a more precise manual classification of the sensor data recorded accordingly. By using classified sensor data and sensor data with a pseudo-classification to train the classifier module, better training of the classifier module may be achieved than would be possible based only on classified sensor data. This may improve the performance of the correspondingly trained classifier module.

[0017] According to one embodiment, the classifier module is configured as a convolutional network.

[0018] This may achieve the technical advantage of providing a powerful classifier module.

[0019] According to one embodiment, classification comprises: [0020] generating latent space representations of the unclassified sensor data in a latent space using the classifier module, wherein latent space representations of sensor data are configured as dimension-reduced representations of the sensor data; [0021] determining distances between latent space representations of the unclassified sensor data and latent space representations of classified sensor data in the latent space using a distance determination module; [0022] classifying the unclassified sensor data with respect to the wall type of the wall and/or the object position and/or the object type of the object based on the distances between the latent space representations of the unclassified sensor data and the latent space representations of the classified sensor data in the latent space, wherein the unclassified sensor data is classified according to a classification of the classified sensor data, if the distances between the latent space representations of the unclassified sensor data and the latent space representations of the classified sensor data in the latent space are less than or equal to a predefined threshold value.

[0023] This may achieve the technical advantage of enabling precise classification of the unclassified sensor data by the classifier module. For this purpose, latent space representations in a latent space are initially generated by the classifier module based on the unclassified sensor data. The latent space representations herein represent dimension-reduced representations of the sensor data. The latent space representations may be configured as vector representations, as is known in the prior art. The latent space representations may in particular be interpreted as dimension-reduced encodings of the information of the sensor data. Further, distances between the latent space representations determined for the unclassified sensor data and latent space representations of already classified sensor data in the latent space are determined by a distance determination module. Based on the determined distances, classification of the unclassified sensor data may subsequently be carried out. Thus, the unclassified sensor data is classified according to classified sensor data, i.e. assigned to the same class of objects as the classified sensor data, if the distances between the latent space representations of the unclassified sensor data and the latent space

representations of the classified sensor data are less than or equal to a predefined threshold value. Classification via distance determination for the latent space representations in the latent space is a precise and reliable method of classifying unclassified sensor data.

[0024] The respective classification or the corresponding object class of the classified sensor data may be considered as an additional entry in the vector representations of the respective latent space representations. Classification by latent space representations is based on the idea that sensor data of the same classification comprises latent space representations with the same or at least very similar vector representations. Such latent space representations have correspondingly small distances from one another in latent space. Classification of unclassified sensor data is thus possible via the distance determination and by selecting a suitable threshold value for the distance.

[0025] According to one embodiment, the distances are defined as Euclidean distances.

[0026] This can achieve the technical advantage of enabling a simple distance determination.

[0027] According to one embodiment, classification comprises: [0028] generating latent space representations of the unclassified sensor data in a latent space using the classifier module, wherein latent space representations of sensor data are configured as dimension-reduced representations of the sensor data; [0029] classifying the unclassified sensor data with respect to the wall type of the wall and/or the object position and/or the object type of the object based on the latent space representations of the unclassified sensor data using a classifier module, wherein the classifier module is configured as an artificial intelligence and trained on latent space representations of classified sensor data to determine a classification of the respective sensor data.

[0030] This may achieve the technical advantage of enabling a precise classification of the unclassified sensor data based on the latent space representations of the unclassified sensor data.

[0031] According to one embodiment, the classifier module is configured as a Generative Adversarial Network GAN, particularly as an autoencoder having an encoder module and a decoder module.

[0032] This may achieve the technical advantage of providing a powerful and reliable classifier module.

[0033] According to one embodiment, the sensor data comprises radar data of a radar sensor and/or data of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an alternating current sensor and/or an NMR sensor and/or an ultrasonic sensor.

[0034] This may achieve the technical advantage that a wide range of different physical metrics can be considered via the different sensor types. This enables the generation of a comprehensive training dataset. Further, this allows for comprehensive training of the artificial intelligence that can be trained to take into account the various physical metrics and the information based on these relating to the wall and the objects disposed therein in the wall diagnosis.

[0035] According to one embodiment, the classification of the sensor data is further performed with respect to an object depth and/or an object extent of the object.

[0036] This may achieve the technical advantage that additional information regarding the properties of the objects disposed in the wall can be incorporated into the training and wall diagnostics. This may improve the wall diagnostics of the diagnostic module.

[0037] According to one embodiment, object classes of the object type of the object comprise: Metal/non-metal object, low voltage cable, single phase AC signal cable, multi phase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, and/or wherein the wall type classes of the wall type of the wall comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, floor heating, wall heating.

[0038] The technical advantage may thereby be achieved that a large number of different objects of different object types can be detected or classified. The measuring device or the diagnostic module can be trained hereby to detect and classify common objects installed in building walls. This allows particularly precise wall diagnostics, in which the detected objects can be precisely and

unambiguously assigned to the corresponding object classes.

[0039] By precisely classifying objects and providing the corresponding classification information to the user via the display unit, wall diagnostics are made as meaningful as possible. Since the user not only knows that an object is located within the wall and where it is, but also which object type the detected object is, the user can decide accordingly how to carry out further work on the wall with respect to the detected object. Providing the object types of the object classification of the detected objects thus represents an essential area of the wall diagnostics, because the user can adjust the planned work on the wall based on the indicated object type accordingly.

[0040] According to one aspect, a training dataset for training an artificial intelligence of a wall diagnostic measuring device is provided, wherein the training dataset was generated by the method for generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments.

[0041] Provided according to one aspect is a computing unit configured to perform the method of generating a training dataset for training an artificial intelligence for operating a measuring device according to one of the preceding embodiments.

[0042] Provided according one aspect is a computer program product comprising instructions that, when the program is executed by a data processing unit, cause the latter to perform the method for generating a training dataset to train an artificial intelligence to operate a measuring device according to one of the preceding embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] Embodiments of the disclosure are described with reference to the following figures. The figures show:

[0044] FIG. 1 a schematic representation of a measuring device according to one embodiment;

[0045] FIG. 2 a further schematic representation of the measuring device according to a further embodiment;

[0046] FIG. 3 a further schematic representation of the measuring device according to a further embodiment;

[0047] FIG. 4 a schematic representation of a measurement of the measuring device according to one embodiment,

[0048] FIG. 5 a further schematic representation of the measuring device according to a further embodiment;

[0049] FIG. 6 a schematic representation of a system for generating a training dataset according to one embodiment,

[0050] FIG. 7 a further schematic representation of a system for generating a training dataset according to a further embodiment,

[0051] FIG. 8 a further schematic representation of the system for generating a training dataset according to a further embodiment,

[0052] FIG. 9 a further schematic representation of the system for generating a training dataset according to a further embodiment,

[0053] FIG. 10 a flowchart of a method for generating a training dataset according to one embodiment,

[0054] FIG. 11 a further flowchart of the method for generating a training dataset according to a further embodiment,

[0055] FIG. 12 a further flowchart of the method for generating a training dataset according to a further embodiment, and

[0056] FIG. 13 a schematic representation of a computer program product.

DETAILED DESCRIPTION

[0057] FIG. 1 shows a schematic representation of a measuring device **100** according to one embodiment.

[0058] The present disclosure relates to a measuring device, in particular a wall diagnostic device for examining walls **105** to be worked on. Wall diagnostic devices are known from the prior art that are used to detect objects located in walls. Such devices allow a user to examine walls to be worked on to search for objects located in the walls, in order to be able to perform planned work, for example drilling in walls, such that damage to the objects located in the walls can be avoided.

[0059] In the embodiment shown, the measuring device **100** comprises a housing **150** having a handle **152** for grasping of the measuring device **100** by a user, a display unit **111** for displaying diagnostic results **109** of the wall diagnostics, and controls **154** for switching the measuring device **100** between various operating modes.

[0060] According to the disclosure, the measuring device **100** comprises at least one radar sensor unit **101**. By way of the radar sensor unit **101**, radar signals may be transmitted towards the wall **105** to be examined, and radar signals reflected by the wall **105** may be received.

[0061] For example, the radar sensor unit **101** may be configured as a narrow band radar detector device in the 2.4 GHz to 2.4835 GHz frequency range or as an ultra-wide band radar detector device in the 1.8 GHz to 5.8 GHz frequency range.

[0062] The measuring device **100** further comprises a diagnostic module **107** executable on a computing unit **151** of the measuring device **100** to perform the wall diagnostics. The diagnostic module **107** is configured to perform corresponding diagnostics on the wall to be examined based on the radar data **103** of the radar sensor unit **101**. The radar data **103** of the radar sensor unit **101** thereby depicts the wall **105** to be examined and, if applicable, objects **113** located within the wall **105**.

[0063] The wall diagnostics carried out by the diagnostic module **107** comprise at least performing object recognition. The object recognition comprises an object detection and an object classification of the object **113** located in the wall **105**. The object detection comprises at least the determination of an object position **115**. The object position describes the positioning of the object located in the wall **105** with respect to a reference system defined by the measuring device **100**. The object classification of the detected object **113** comprises at least determining an object type **117** of the detected object **113**.

[0064] The diagnostic results of the wall diagnostics determined in this way, i.e. at least the determined object position **115** and/or the determined object type **117** of the object **113** located in the wall **105**, are subsequently presented in a display unit **111** of the measuring device **100** to a user of the measuring device **100**. For example, the display unit **111** may be configured as a corresponding display, and the diagnostic results **109** may be visually displayed. Additionally, the display of the diagnostic results **109** may be supported via audible and/or haptic signals. For example, the haptic signals may be realized via corresponding vibration signals.

[0065] The object **113** may be shown in the display, for example, by way of a corresponding icon. The object **113** can be shown in the corresponding object position **115** in the display. The object extension **121** may be visualized by a corresponding size of the displayed icon. The particular object type **117** of the object **113** may be visualized with a corresponding term or color highlighting of the icon, or by a specific shape of the icon representing the object **113**.

[0066] Alternatively, the wall diagnostics may additionally comprise determining a wall type **123** in the form of a wall type classification of the wall **105** to be examined. The wall type **123** describes the respective type of the wall **105** to be examined. For example, the wall type may be associated with corresponding wall type classes, which may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or wall bricks, underfloor heating, wall heating or similar wall types found in buildings.

[0067] According to one embodiment, the diagnostic module **107** is further configured to determine

an object depth **119** of the object **113** within the wall **105** based on the radar data **103**. The object depth **119** is defined by a distance of the object formed in the wall **105** to a surface of the wall **105**. The distance may be defined on the object side, for example with respect to an object surface or with respect to an object center point. The distance to the surface of the wall **105** describes a shortest distance defined by a direction perpendicular to the surface of the wall **105**.

[0068] According to one embodiment, the diagnostic module **107** is further configured to determine an object extension **121** of the object **113** in at least one predefined direction based on the radar data **103**. The object extension **121** of the object **113** describes a spatial extension of the object **113** in at least one spatial direction, preferably in two spatial directions, particularly preferably in three spatial directions. The object **113** may be described here as a one-dimensional, two-dimensional, or three-dimensional object **113**.

[0069] In conventional use, the measuring device **100** is placed on the surface of the wall **105** to be examined. Radar signals are transmitted towards the wall **105**, and radar signals reflected from the wall **105** or the objects **113** located behind it are received, via the radar sensor unit **101**. This radar data **103** of the radar sensor unit **101** is used by the diagnostic module **107** to perform the wall diagnostics described above and to determine corresponding diagnostic results **109**.

[0070] For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** of the object **113** located in the wall **105**. Alternatively or additionally, the diagnostic results **109** may comprise the wall type **123** of the wall **105** and/or the object depth **119** and/or the object extension **121** of the object **113**.

[0071] The diagnostic results **109** configured in this manner may subsequently be displayed to a user of the measuring device **100** in a display unit **111** of the measuring device **100**. The display unit **111** can be configured as a corresponding display, for example. The diagnostic results **109** may be displayed in graphical or textual form in the display unit **111**.

[0072] According to one embodiment, the measuring device **100** further comprises a motion detection unit **141**. The motion detection unit **141** may be used to detect movement of the measuring device **100** relative to the wall **105**. The motion detection unit **141** may comprise, for example, at least one roller element for this purpose. When the roller element is placed on the wall surface of the wall **105**, movement of the measuring device **100** relative to the wall **105** can be detected when the measuring device **100** moves along a direction of movement **153** by rolling the roller element. Alternatively, the motion detection unit **141** may have a different configuration by which a relative movement of the measuring device **100** relative to the wall **105** can be detected.

[0073] By moving the measuring device **100** relative to the wall **105**, radar data **103** of the radar sensor unit **101** may be captured for a plurality of different positions of the measuring device **100** relative to the wall **105**. This allows a larger spatial area of the wall **105** to be examined than the effective range of the radar sensor unit **101**. This allows for objects **113** to be captured that have a greater spatial extent than the effective range of the radar sensor unit **101**.

[0074] While the measuring device **100** moves along the direction of movement **153**, radar data **103** of the radar sensor unit **101** may be captured continuously. The wall diagnostics may be evaluated by the diagnostic module **107** based on this radar data **103** while the measuring device **100** is moving along the direction of movement **153**. This allows for accelerated wall diagnostics, taking into account the positioning of the measuring device **100** relative to the wall **105**.

[0075] According to its embodiment, the diagnostic module **107** is configured as a correspondingly trained artificial intelligence **125**. The artificial intelligence **125** is trained to perform the above-mentioned wall diagnostics based on the radar data **103** of the radar sensor unit **101** and to determine at least the object position **115** and the object type **117** of an object **113** located in the wall **105**. The object classification or determination of the object type **117**, respectively, comprises assigning the detected object **113** to predefined object classes.

[0076] The object classes may comprise: Metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water

filled plastic pipe, for example fresh water pipe, non-water filled plastic pipe, for example waste water pipe, or other elements commonly installed in building walls.

[0077] Furthermore, the artificial intelligence **125** may be trained to determine the wall type **123** of the wall **105** to be examined at least based on the radar data **103** of the radar sensor unit **101**.

Possible wall types **123** may comprise: Concrete wall, plasterboard/drywall wall, brick wall and/or individual bricks of the brick wall, underfloor heating, wall heating or other similar wall types commonly installed in buildings.

[0078] According to one embodiment, in addition to the radar sensor unit **101**, the measuring device **100** may comprise further additional sensors by way of which additional physical variables are detectable. For example, the measuring device **100** may comprise an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an AC current sensor and/or an NMR sensor and/or an ultrasonic sensor or other sensors commonly used in wall diagnostic devices.

[0079] The diagnostic module **107**, in particular the corresponding trained artificial intelligence **125**, can be configured to perform wall diagnostics based on the radar data **103** of the radar sensor unit **101** and taking into account the additional sensor information of the further sensors described above. The additional information of the additional sensors mentioned above can particularly be used for object recognition of the objects **113** located in the walls **105**. The additional sensor information may possibly provide improved detection of the objects **113** and may possibly provide improved classification of the objects **113**.

[0080] For example, the material of the objects **113**, for example as a metallic or non-metallic material, can be particularly improved and classified by using the additional sensor information.

[0081] FIG. 2 shows another schematic representation of the measuring device **100** according to a further embodiment.

[0082] In the embodiment shown, the measuring device **100** comprises a pre-processing module **127** in addition to the diagnostic module **107**. For wall diagnostics, the measuring device **100** first receives the radar data **103** of the radar sensor unit **101**. Pre-processing of the receiving radar data **103** is performed via the pre-processing module **127**. For example, via pre-processing by the pre-processing module **127**, the radar data may be brought into a corresponding data structure required for wall diagnostics by the diagnostic module **107**.

[0083] As described above, during wall diagnostics, the diagnostic module **107** generates the diagnostic results **109** described above. For example, the diagnostic results **109** may comprise the object position **115** and/or object type **117** and/or object depth **119** and/or object extension **121** of an object **113** formed in the wall **105** to be examined and/or the wall type **123** of the wall **105** to be examined. The correspondingly generated diagnostic results **109** may subsequently be displayed in the display unit **111** of the measuring device **100**.

[0084] According to one embodiment, in addition to the radar data **103** of the radar sensor unit **101**, the additional sensor information of the additional sensors described above may be considered during the wall diagnostics of diagnostic module **107**. A corresponding pre-processing of the additional sensor information may be performed accordingly by the pre-processing module **127**.

[0085] In the embodiment shown, the diagnostic module **107** comprises a wall type classification module **129** and an object recognition module **131**. The pre-processing module **127** comprises a first pre-processing module **135** and a second pre-processing module **137**. The first pre-processing module **135** comprises an S matrix reduction **155**. The second pre-processing module **137** comprises a background correction **157**, an inverse Fast Fourier transformation **159**, and a focusing and migration **161**. During the pre-processing of the radar data **103** by the pre-processing module **127**, the radar data **103** is first pre-processed by the first pre-processing module **135** and the S-matrix reduction **155** contained therein.

[0086] When doing so, the first pre-processing module **135** generates input data **133** based on the radar data **103**. The input data **133** serves as input data for the wall type classification module **129**. The wall type classification module **129** performs a wall type classification of the wall **105** to be

examined based on the input data **133** and generates wall type information **139**. The wall type information **139** contains the wall type **123** of the wall **105** to be examined, as determined in the wall type classification.

[0087] Subsequently, the second pre-processing module **137** performs pre-processing based on the radar data **103** and the wall type information **139**. A background correction **157** of the radar data **103** is performed during this, taking into account the wall type **123** determined in the wall type information **139**. Depending on the wall type **123** of the wall **105** to be examined, different effects can occur on the radar data **103**.

[0088] These effects, which are primarily based on the respective wall type **123** and can affect object recognition, can be corrected by the background correction **157**. After the background correction has been performed, further pre-processing can be carried out by performing the inverse Fast Fourier transformation **159** or focusing and migration **161**, respectively, and input data **133** can be created for the object recognition module **131** once again. Based on the input data **133** provided by the second pre-processing module **137**, the object recognition module **131** performs object recognition of the object **113** located in the wall **105** to be examined and determines at least the object position **115** and the object type **117** of the respective object **113**. Additionally, the object depth **119** and the object extension **121** may be determined by the object recognition module **131**.

[0089] According to one embodiment, the diagnostic module is further configured to determine an object depth of the object within the wall based on the radar data, wherein the object depth is defined by a distance of the object formed in the wall to a surface of the wall.

[0090] The pre-processing is optional. Depending on the algorithm used for the diagnostic module **107**, completely unprocessed radar echoes of different frequencies may be used as radar data **103** and as input data for the diagnostic module **107**. Alternatively, radar data **103** processed via multiple steps may be used. The pre-processing steps comprise, for example, the transformation of the signals from the frequency domain to the time or distance domain, background deduction, noise removal, and normalization of the signals. For radar data **103** which is available in the form of complex numbers, only the absolute value can be processed. Alternatively or additionally, the phase information may be considered.

[0091] FIG. **3** shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0092] In the embodiment shown, the diagnostic module **107** comprises a plurality of parallel processing paths **102**. Each processing path **102** includes a pre-processing module **127**, the diagnostic module **107**, comprising the wall type classification module **129** and/or the object recognition module **131**, for example, according to the embodiment in FIG. **2**, and a post-processing module **163**.

[0093] In FIG. **3**, primarily the radar data **103** is shown as input data for the wall diagnostics. In addition to the radar data shown, however, the additional information from the additional sensors may also serve as input data for the wall diagnostics. The different information from the different types of sensors in the different parallel processing paths **102** can be processed and the corresponding wall diagnostics performed separately on the different sensor information. Upon completion of the wall diagnostics, a summary module may be used to assemble a summary of the individual partial analysis results for the diagnostic results **109** of the wall diagnostics.

[0094] Alternatively or additionally, different sub-aspects of wall diagnostics may be performed by the different processing paths **102** based on the same sensor information.

[0095] For example, the individual processing paths **102** may process different radar data **103** captured during movement of the measuring device **100** relative to the wall **105** for different positions of the measuring device **100** relative to the wall **105**. The radar data **103**, thus representing different regions of the wall **105** and captured sequentially in time as the measuring device **100** moves relative to the wall **105**, may then be processed in the different processing paths **102** by the modules shown.

[0096] The various processing paths perform stand-alone wall diagnostics including at least determining the object position **115** and/or the object type **117** of the object **113** located in the wall **105**.

[0097] By way of the summary module **165**, the partial results of the stand-alone wall diagnostics of the different regions of the wall **105** provided in the individual processing paths **102** can be summarized to a contiguous diagnostic result **109**. The contiguous diagnostic result describes the wall diagnostics of a contiguous spatial area that was scanned during movement of the measuring device **100** relative to the wall **105** and depicted by the corresponding captured radar data **103**. The parallel processing of the radar data **103** or additional sensor information **104** of the additional sensor elements in the different processing paths **102** thus enables accelerated wall diagnostics.

[0098] Alternatively, various wall diagnostic functions may also be performed in the different processing paths **102**. For example, in one processing path **102**, the wall type classification and the determination of the wall type **123** of the wall **105** to be examined can be performed. In another processing path **102**, object recognition of the object **113** located in the wall can be performed. The object detection can be performed with the determination of the object position **115** and the object classification can be performed with the determination of the object type **113** in one processing path **102**.

[0099] Alternatively, the object detection and object classification may then also be performed in two separate processing paths **102**. In further processing paths **102**, the object depth determination, i.e., the determination of the object depth **119** and/or the determination of the object extension **121** may each be carried out. In the summary module **165**, the various partial results of the wall diagnostics may be summarized into corresponding diagnostic results **109**.

[0100] The diagnostic module **107** can be divided into different artificial intelligence structures **125**, as already shown in the embodiment in FIG. 2. For example, the diagnostic module **107** may comprise a wall type classification module **129** and an object recognition module **131**. The object recognition module may in turn be divided into an object detection module and an object classification module. The diagnostic module **107** may further comprise an object depth determination module and object extension module, respectively configured to determine the object depth **119** and the object extension **121**.

[0101] The respective modules may each be configured as stand-alone artificial intelligence structures **125**, for example, neural networks. Alternatively, the various modules may form portions of an overall artificial neural network that are connected to an overall neural network according to structures known from prior art.

[0102] FIG. 4 shows a schematic illustration of a measurement by the measuring device **100** according to one embodiment.

[0103] For pre-processing, the radar data **103** or the additional sensor information **104** of the remaining sensors may be normalized, in particular to numerically stabilize the subsequent steps performed by the diagnostic module **107** during the wall diagnostics. For example, an amplitude and/or offset compensation may be performed for this purpose. Moreover, to reduce interference, filtering of the radar data **103** may be carried out, and to reduce the data rate, the corresponding sensor data may be sampled. Additionally, the radar data **103** or the additional sensor information **104** may be transformed into the respectively required frequency range or time range. Methods known from the prior art can be used for this purpose.

[0104] Furthermore, the captured radar data **103** or additional sensor information **104** may be divided into temporal or spatial windows **167**. Temporal windows **167** may be generated by recording the radar data **103** or the additional sensor information or the pre-processed radar data **103** over a fixed time interval. Spatial windows **167**, on the other hand, may be generated from a mapping of the radar data **103** or additional sensor information **104** to positions of the measuring device **100** relative to the wall **105** along the direction of movement **153**.

[0105] Graph a) of FIG. 4 shows such a data matrix resulting from the steps described above. The

data matrix of the window **167** shown in graph a) shows a plurality of sensor data, which may comprise, for example, radar data **103** or additional sensor information **104** of the further sensors plotted along a frequency channel axis **171** or along a space/time axis **169**, respectively.

[0106] A width of the temporal windows **167** may be selected such that different sampling rates of the sensors may be balanced and a new window **167** may be provided frequently enough so that a display of the diagnostic results **109** of the wall diagnostics in the display unit **111** may be shown without too great of a time delay while the measurement is being performed or shortly after the measurement by the measuring device **100** ends.

[0107] A rate of 2 to 20 windows per second of the data recording of the sensor data may be advantageous for this purpose. For spatial windows, the spatial sampling rates can be selected to achieve the desired local accuracy. Advantageously, 1 mm to 1 cm sampling rates can be used. This means that corresponding sensor data is captured every 1 mm to 1 cm of movement of the measuring device **100** along the direction of movement **153**.

[0108] A width of the spatial windows **167** may be selected such that contiguous information relating to an object **113** is contained in a window. Advantageously, a width of the respective spatial windows **167** can be 1 cm to 20 cm. This results in 4 to 100 measured values per window **167**. This allows for further efficient algorithmic processing of the correspondingly recorded radar data **103** or additional sensor information by the diagnostic module **107**.

[0109] A further temporal window **167** or spatial window **167** can be provided as soon as one or more sampling points are available.

[0110] The diagnostic module **107** may be oriented such that a matrix corresponding to the window size of the respective spatial or temporal window **167** may be included as input data, for example of each processing path **102** of the embodiment in FIG. 3 as well. According to the embodiment of FIG. 2, the corresponding input data can comprise the respective pre-processed sensor data, i.e. radar data **103** and additional sensor information **104** of the additional sensors.

[0111] As stated above, the wall diagnostics may be performed by the diagnostic module **107** based on a correspondingly trained artificial intelligence. Alternatively, different processing paths may also be calculated by way of rule-based algorithms. In addition, within a processing path **102**, a combination of artificial intelligence and rule-based algorithm is possible in the form of a parallel operation or concatenation.

[0112] The diagnostic results **109** of the wall diagnostics may be expressed as numeric values, vectors, or matrices. Furthermore, for the object detection, the probability of detection, or for the wall type classification and/or the object classification, respectively, a probability of the specified object classes and/or wall type classifications can be indicated. The same may apply to the position and/or depth determination, for which corresponding probability values can also be indicated.

[0113] In addition to the radar data **103**, if the additional sensor information of the further sensor types is processed in a processing path **102**, these may either be merged within the artificial intelligence **125** or combined by way of rule-based combinations.

[0114] During the post-processing of each processing path **102**, of the embodiment in FIG. 3, multiple algorithm results based on multiple windows **167** may be summarized by the summary module **165**. This summary can in particular be realized by way of majority formation, sum formation or also by multiplication of successive probability values.

[0115] Furthermore, by clustering multiple results, for example, it is possible to detect which objects, of multiple detected objects located close to one another, are the same object so that they are not incorrectly detected multiple times.

[0116] Likewise, it is possible to multiplicatively apply a weighting function **177** when summarizing the results from multiple windows **167**. Advantageously, the diagnostic partial results **175** corresponding to corresponding data points in the space may be weighted with respect to positioning of the diagnostic partial results **175** relative to a center point of the respective window **167**. This is illustrated by way of example in graph b), in which the individual diagnostic partial

results **175** are weighted according to the weighting function **177** shown with respect to the center point of the window **167** shown.

[0117] According to one embodiment, the results of one processing path **102** after post-processing **163** may influence the extension of another processing path **102**. Weighting parameters may be adjusted that may depend on the particular result from the processing path **102** for each window.

[0118] For example, the result of an object classification in which the object type **117** of an object located in the wall **105** is defined, can be used to increase the weight of a wall type classification, in which the wall type **123** of the respective wall **105** is determined, during the post-processing at locations without objects **113**, because the respective radar data **103** at these locations is less influenced by reflections of the objects **113**.

[0119] FIG. 5 shows a further schematic representation of the measuring device **100** according to a further embodiment.

[0120] Graphs a) and b) of FIG. 5 show two different alternatives of joint data processing of radar data **103** and additional sensor information **104** by the diagnostic module **107**.

[0121] Graph b) illustrates a joint processing of the radar data **103** and the additional sensor information **104** of the additional sensors by the diagnostic module **107**. For this purpose, the radar data **103** and the additional sensor information **104** are collectively used as input data of the diagnostic module **107** configured as an artificial intelligence, in particular as an artificial neural network. The diagnostic module **107** here comprises multiple convolutions **108** and multiple dense layers **106**. The radar data **103** and the additional sensor information **104** are processed jointly as input data via the convolutions **108** and dense layers **106**. The aforementioned diagnostic results **109** are created as output data of the diagnostic module **107** based on this.

[0122] On the other hand, in graph b) the radar data **103** and the additional sensor information **104** are used as stand-alone input data of the diagnostic module **107**. The diagnostic module **107** becomes multiple processing paths **102**. The processing paths **102** each comprise multiple convolutions **108** and at least one dense layer **106**. In the various processing paths **102**, wall diagnostics are performed separately by the diagnostic module **107** based on the radar data **103** and the additional sensor information **104**, respectively.

[0123] In an additional concatenation layer **148**, the partial results of the partial diagnostics of the different processing paths **102** are combined and fed to a final dense layer **106**. The output data of the diagnostic module **107** corresponds to the diagnostic results **109** described above.

[0124] The correspondingly configured diagnostic module **107** is configured to perform wall diagnostics as described above, including the features described above, based on the radar data **103** and the additional sensor information **104**.

[0125] In the embodiment shown, the diagnostic module **107** is configured as an artificial neural network, in particular as a convolutional network. Corresponding network architectures with convolutions **108**, dense layers **106**, and concatenation layers **148** are known from the prior art.

[0126] FIG. 6 shows a schematic representation of a system **600** for generating a training dataset **143** according to one embodiment.

[0127] In the embodiment shown, the system **600** for generating a training dataset **143** comprises classifier module **183**. The classifier module **183** is configured as an artificial intelligence trained to generate classified sensor data **174** based on unclassified sensor data **172**.

[0128] The unclassified sensor data **172** here depicts walls **105** to be examined and objects **113** disposed therein. The unclassified sensor data **172** may include radar data **103** or additional sensor information **104**. The unclassified sensor data **172** may be based on a plurality of measurements of a plurality of measuring devices **100** of a plurality of different walls **105**.

[0129] According to the disclosure, the classifier module **183** is trained via a semi-supervised learning process to determine a classification for unclassified sensor data in relation to the wall type **123** of the wall **105** and/or the object position **115** and/or the object type **117** of the object **113**.

[0130] A corresponding training dataset **143** may subsequently be generated by the system **600**

based on the sensor data **174** classified by the classifier module **183**.

[0131] The classifier module **183** may be further trained to classify the unclassified sensor data **172** in addition to wall type **123**, object position **115**, and/or object type **117** with respect to the object depth **119** and/or object extent **121**

[0132] FIG. **7** shows a further schematic representation of the system **600** for generating a training dataset **143** according to a further embodiment.

[0133] The embodiment in FIG. **7** is based on the embodiment in FIG. **6** and includes all the features described there.

[0134] Graphic a) illustrates a training process of the classifier module **183**. On the other hand, graphic b) represents the actual application process of the classifier module **183** trained according to the training method of graphic a).

[0135] To train the classifier module **183**, a training dataset is generated based on classified sensor data **174** and sensor data having a pseudo-classification **176**. The classifier module **183** is subsequently trained based on the classified sensor data **174** and the sensor data with a pseudo-classification **176** to classify unclassified sensor data **172** and generate classified sensor data **174**.

[0136] The sensor data with a pseudo-classification **176** is generated by a pseudo-classification module **184**. The pseudo-classification module **184** is configured as a trained artificial intelligence trained to generate sensor data with a pseudo-classification **176** based on unclassified sensor data **172**. The sensor data with a pseudo-classification **176** is to be understood here as classified sensor data generated by the pseudo-classification module **184**.

[0137] The pseudo-classification module **184**, in turn, has been trained on classified sensor data **174** to generate sensor data with a pseudo-classification **176** based on unclassified sensor data **172**.

[0138] The classifier module **183** is subsequently classified based on the classified sensor data **174** and the sensor data generated by the pseudo-classification module **184** with a pseudo-classification **176**.

[0139] The classified sensor data **174** of the graphic a) used for training the pseudo-classification module **184** and the classifier module **183** does not correspond to the classified sensor data **174** of the graphic b) generated by the trained classifier module **183** based on the unclassified sensor data **172**. The classified sensor data **174** of the training shown in graphic a) are, on the other hand, manually classified sensor data that were classified in a classification process by an experimenter.

[0140] The unclassified sensor data **172** used during this in graphic a) to generate the sensor data with a pseudo-classification **176** also does not correspond to the unclassified sensor data **172** used during application of the trained classifier module **183** to generate the training dataset.

[0141] The unclassified sensor data **172** of the graphics a) and b) may be identical, but does not have to be.

[0142] FIG. **8** shows a further schematic representation of the system **600** for generating a training dataset **143** according to a further embodiment.

[0143] In the embodiment shown, the classifier module **183** is configured as an autoencoder having an encoder module **185** and a decoder module **187**. Via the encoder module, the classifier module **183** is configured to generate corresponding latent space representations **189** based on sensor data **173**. The latent space representations **189** correspond to dimension-reduced representations of the sensor data **173**. In the prior art of machine learning, such latent space representations are configured as vector representations. The sensor data **173** may include unclassified and/or classified sensor data **172**, **174**.

[0144] The decoder module **187**, on the other hand, is configured to decode the information encoded in latent space representations **189** of the sensor data **173** and to generate sensor data **191** generated based on the latent space representations **189**. Thus, new sensor data may be generated via the decoder module **187** based solely on latent space representations **189** and not generated by measurements of corresponding sensor units.

[0145] To classify unclassified sensor data **172** and generate classified sensor data **174**, the

classifier module **183** further comprises a distance determination module **193**. Distances between the latent space representations **189** in latent space **195** can be determined via the distance determination module **193**.

[0146] Graphic b) shows such classification of unclassified sensor data **172** via a distance determination of corresponding latent space representations in latent space **195**.

[0147] First, the correspondingly trained encoder module **185** determines a corresponding latent space representation **188** of the unclassified sensor data **172** for the unclassified sensor data **172**. In graphic b), three groups of latent space representations **190** of classified sensor data **174** are further depicted in the latent space **195**. The latent space representations **190** are summarized into three groups spaced apart from each other. The latent space representations **190** of the three different groups each represent classified sensor data **174**. The latent space representations **190** of a common group here represent sensor data of a common object class. To classify the unclassified sensor data **172**, the distance determination module **193** subsequently determines distances D between the latent space representations **188** of the unclassified sensor data **172** and the latent space representations **190** of the classified sensor data **174**. Now, if the determined distance D between the latent space representations **188** and the latent space representations **190** of the classified sensor data **174** is less than or equal to a predetermined threshold, then the respective unclassified sensor data **172** is assigned the respective classification of the classified sensor data **174** for which the distance D of the respective latent space representation **188** is less than or equal to the predetermined threshold.

[0148] Because the classifications of the classified sensor data **174** are already known, they may be incorporated into the respective latent space representations **190** as additional information. In this way, the distance determination may then be performed by the distance determination module **193** for the various latent space representations **188**, **190** in the latent space **195**, for example, by a Euclidean distance metric. Latent space representations **190** representing classified sensor data **174** of the same classification are spaced close together because they have the same classification information in latent space and are arranged in the groups shown in graphic B. Thus, the classification of the unclassified sensor data **172** can be carried out via the distance determination D .

[0149] Graphic b) shows such a classification. The latent space representation **188** of the unclassified sensor data **172** is spaced apart from the latent space representations **190** of the classified sensor data **174** shown in squares at a distance D that is less than the predefined threshold value. This is characterized in that the latent space representation **188**, marked X, is disposed in the group of latent space representations **190**, marked with square icons.

[0150] Latent space representations **188** which, on the other hand, have a greater distance D than the predefined threshold value to the latent space representations **190** of the classified sensor data **174** cannot be classified via the method shown. For example, such unclassified sensor data **172** may be disregarded for the training dataset **143** to be generated.

[0151] FIG. **9** shows a further schematic representation of the system **600** for generating a training dataset **143** according to a further embodiment.

[0152] The embodiment shown in FIG. **9** is based on the embodiment in FIG. **6** and includes all the features described there. In the embodiment shown, the classifier module **183** comprises an encoder module **185** trained according to the embodiment in FIG. **8**. The encoder module **185** is configured to generate latent space representations **189** based on sensor data **173**. The sensor data **173** may comprise unclassified and classified sensor data **172**, **174**. Further, the classifier module **183** includes a classification module **192**.

[0153] Thus, to classify unclassified sensor data **172**, the correspondingly trained encoder module **185** generates latent space representations **188** of the unclassified sensor data **172**.

[0154] The classifier module **192** may generate correspondingly classified sensor data **174** based on the latent space representations **188** of the unclassified sensor data **172**. The classifier module

192 is configured as an artificial intelligence and trained to predict the corresponding classification of the sensor data based on latent space representations **188** of unclassified sensor data. The classifier module **192** may have been trained on latent space representations **190** of classified sensor data **174** to predict corresponding classifications based on latent space representations **188** of unclassified sensor data **172**.

[0155] FIG. **10** shows a flowchart of a method **300** for generating a training dataset **143** according to one embodiment.

[0156] To generate a training dataset, in a first method step **301**, unclassified sensor data **172** of at least one sensor unit **101** of the measuring device **100** is first received by the classifier module **183**.

[0157] In a further method step **303**, the unclassified sensor data **172** is classified and classified sensor data **174** is provided by the classifier module **183**. The classifier module **183** is configured as an artificial intelligence and trained via semi-supervised learning to determine a classification for unclassified sensor data **172** with respect to the wall type **123** of the wall **105** and/or the object position **115** and/or the object type **117** of the object **113**.

[0158] In a further method step **305**, the sensor data **174** classified in this manner is aggregated into a training dataset **143** or added to an already existing training dataset **143**.

[0159] According to one embodiment, the classification may be carried out, in addition to the wall type **123**, the object position **115**, and/or object type **117**, with respect to the object depth **119** and/or object extent **121**.

[0160] FIG. **11** shows a further flowchart of the method **300** for generating a training dataset **143** according to a further embodiment.

[0161] The embodiment shown in FIG. **11** is based on the embodiment in FIG. **10** and comprises all the method steps described there.

[0162] In the embodiment shown, latent space representations **188** are generated by the classifier module **183** for the unclassified sensor data **172** for classification **303** in a method step **307**.

[0163] In a further method step **309**, distances **D** between the latent space representations **188** of the unclassified sensor data **172** and latent space representations **190** of classified sensor data **174** are determined by the distance determination module **193**.

[0164] In a further method step **311**, the unclassified sensor data **172** is classified with respect to the wall type **123** and/or the object position **115** and/or the object type **117** based on the distances **D** between the latent space representations **188** of the unclassified sensor data **172** and the latent space representations **190** of the classified sensor data **174**. The unclassified sensor data **172** is classified according to a classification of the classified sensor data **174**, if the distances **D** between the latent space representations **188** of the unclassified sensor data **172** and the latent space representations **190** of the classified sensor data **174** are less than or equal to a predefined threshold value.

[0165] FIG. **12** shows a further flowchart of the method **300** for generating a training dataset **143** according to a further embodiment.

[0166] The embodiment in FIG. **12** is based on the embodiment in FIG. **10** and includes all the features described there.

[0167] In the embodiment shown, the latent space representations **188** of the unclassified sensor data **172** are generated to classify **303** the unclassified sensor data **172** in method step **307**. In a further method step **313**, the unclassified sensor data **172** is classified with respect to the wall type **123** and/or the object position **11** and/or the object type **117** based on the latent space representations **188** of the unclassified sensor data **172** by the classifier module **192**. The classifier module **192** is configured as an artificial intelligence and is trained to determine corresponding classifications of the sensor data based on latent space representations **188** of unclassified sensor data **172**.

[0168] FIG. **13** shows a schematic representation of a computer program product **500** comprising instructions that, when the program is executed by a data processing unit, cause the latter to

perform the method **300** for generating a training dataset **143**.

[0169] In the embodiment shown, the computer program product **500** is stored on a storage medium **501**. The storage medium **501** can in this case be any desired storage medium known from the prior art.

Claims

1. A computer-implemented method for generating a training dataset for training an artificial intelligence for operating a measuring device, comprising: receiving unclassified sensor data of at least one sensor unit of a measuring device using a classifier module, wherein the unclassified sensor data depicts a wall to be diagnosed with a wall type, and wherein an object of an object type is disposed in the wall in an object position; classifying the unclassified sensor data and providing classified sensor data using a classifier module, wherein the classifier module is configured as an artificial intelligence and trained via semi-supervised learning to determine a classification for unclassified sensor data with respect to the wall type of the wall and/or the object position and/or the object type of the object; and adding the classified sensor data to a training dataset.
2. The method of claim 1, wherein the classifier module is trained based on classified sensor data and sensor data with pseudo-classifications to determine corresponding classifications for unclassified sensor data, wherein the pseudo-classifications were predicted by a pseudo-classification module based on unclassified sensor data, and wherein the pseudo-classification module is configured as an artificial intelligence and is trained based on classified sensor data to determine pseudo-classifications for unclassified sensor data.
3. The method of claim 1, wherein classification comprises: generating latent space representations of the unclassified sensor data in a latent space using the classifier module, wherein latent space representations of unclassified or classified sensor data are formed as dimension-reduced representations of the sensor data; determining distances between the latent space representations of the unclassified sensor data and latent space representations of classified sensor data in the latent space using a distance determination module; and classifying the unclassified sensor data with respect to the wall type of the wall and/or the object position and/or the object type of the object based on the distances between the latent space representations of the unclassified sensor data and the latent space representations of the classified sensor data in the latent space, wherein the unclassified sensor data is classified according to a classification of the classified sensor data, if the distances between the latent space representations of the unclassified sensor data and the latent space representations of the classified sensor data in the latent space are less than or equal to a predefined threshold value.
4. The method of claim 1, wherein classification comprises: generating latent space representations of the unclassified sensor data in a latent space using the classifier module, wherein latent space representations of unclassified or classified sensor data are formed as dimension-reduced representations of the sensor data; and classifying the unclassified sensor data with respect to the wall type of the wall and/or the object position and/or the object type of the object based on the latent space representations of the unclassified sensor data using a classifier module, wherein the classifier module is configured as an artificial intelligence and is trained to determine a classification of the respective sensor data based on latent space representations of classified sensor data.
5. The method of claim 4, wherein the classifier module is configured as a Generative Adversarial Network.
6. The method of claim 1, wherein the sensor data comprises radar data of a radar sensor and/or data of an induction sensor and/or an eddy current sensor and/or a capacitance sensor and/or an alternating current sensor and/or an NMR sensor and/or an ultrasonic sensor.
7. The method of claim 1, wherein the classification of the sensor data is further performed in

relation to an object depth and/or an object extension of the object.

8. The method of claim 1, wherein object classes of the object type of the object comprise: metallic/non-metallic object, low voltage cable, single phase AC signal cable, multiphase AC signal cable, wood beam, metal beam, plastic pipe, water filled plastic pipe, non-water filled plastic pipe, and/or wherein the wall type classes of the wall type of the wall comprise: concrete wall, plasterboard/drywall wall, brick wall and/or bricks of the wall, underfloor heating, wall heating.

9. A training dataset for training an artificial intelligence of a wall diagnostic measuring device, wherein the training dataset was generated according to the method for generating a training dataset according to claim 1.

10. A computing unit configured to perform the method of generating a training dataset for training an artificial intelligence for operating a measuring device of claim 1.

11. A computer program product comprising instructions which, when the program is executed by a data processing unit, prompt the data processing unit to perform the method for generating a training dataset for training an artificial intelligence to operate a measuring device according to claim 1.

12. The method of claim 1, wherein the measuring device is a wall diagnostic device.

13. The method of claim 4, wherein the classifier module is configured as an autoencoder having an encoder module and a decoder module.

14. The method of claim 8, wherein: the water filled plastic pipe is a fresh water pipe, and the non-water filled plastic pipe is a waste water pipe.
